Yang Chen [ORCID: 0009-0007-8814-2730]

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EDUCATION

Sun Yat-sen University Shenzhen, China
School of Intelligent Systems Engineering 09/2021 to present
Bachelor of Intelligent Transportation Engineering expected 06/2025
Cumulative GPA: 3.7/4.0 ; Research Supervisor: Professor Mengtang Li[1]

- ✓ Second-Class Scholarship of Sun Yat-sen University
- ✓ Excellent Student Leader
- ✓ Outstanding award for artistic and athletic performance
- ✓ Meritorious award for morality

PUBLICATION

(Under review) Li, M., Wang, B., ***Chen, Y.*** (Second Author), Zhang, J., & Wei, L.Novel Embedded Magnetic Field Sensor Arrays for Real-time Shape Estimation of Tendon-driven Continuum Robots
Submitted to 2026 IEEE International Conference on Robotics and Automation (ICRA)

RESEARCH/PROJECTS (Selected)

A. Intelligent Transportation Systems, Simulation & Video Analytics

Carbon footprint prediction of residents in the center of a megacity based on multi-source transportation data[2] Spring 2024
Participatory Investigator (Supervisor: Associate Professor Yonghong Liu)

- Built metro/bus travel-chain reconstruction pipeline using IC-card & GPS temporal–spatial matching.
- Modeled individual carbon emissions based on transit energy-consumption factors.
- Developed an interactive visualization tool for mapping travel chains and carbon footprints.

Tools: Multi-source Data Processing, GIS-based visualization ,Python, Pandas, NumPy, Folium

System for the analysis of unusual events at urban crossroads based on people-vehicle non-trajectory perception[3] Spring 2023
Participatory Investigator (Supervisor: Dr. Yiting Zhu; Associate Professor Ronghui Zhang)

- Extracted multimodal trajectories (pedestrians, cyclists, vehicles) from intersection video data.
- Detected abnormal events through trajectory–facility spatiotemporal matching.
- Analyzed safety risks and identified key conflict zones.
- Built SUMO simulations to reproduce events and evaluate mitigation strategies.

Tools: SUMO, Python,OpenCV

SUMO-based Urban Traffic Simulation & Signal Timing (Course) Fall 2022

- Constructed traffic networks using Netedit and OSM; configured vehicle and pedestrian flows.
- Designed and simulated intersection signal plans using Python and deployed them in SUMO.
- Modeled lane connections, public transport stops, turning rules, and pedestrian facilities

Tools: SUMO, Python

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[1] https://www.researchgate.net/profile/Mengtang-Li
[2] Based on data of two public transportation systems (metro and bus) of the city Chengdu, China
[3] Recommended by Sun Yat-sen University for research merit

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Capacity of pedestrian crossing at an intersection and safety research report as well as target detection effectiveness of pedestrian and non-motorized vehicle Fall 2022

Project Leader and Principal Investigator (Supervisor: Associate Professor Zhaocheng He)

- Collected UAV video data and applied YOLOv5 + DeepSort for pedestrian and cyclist detection and tracking.
- Analyzed pedestrian level of service and computed violation rates (e.g., red-light running, illegal riding).
- Identified conflict patterns among pedestrians, cyclists, and vehicles using trajectory-based analysis.
- Reviewed literature and proposed road design improvements to reduce multimodal traffic conflicts.

Tools: Python, YOLOv5, DeepSort, Full-Cycle Traffic Survey and Analysis

B. Computer Vision, Deep Learning & 3D Reconstruction

Optimization and function expansion of a facial expression recognition system Fall 2023

Participatory Investigator (Supervisor: Associate Professor Chao Gou)

- Built a PCA + KNN baseline and optimized dataset preprocessing.
- Developed a CNN-based expression recognition model with improved accuracy.
- Implemented a CNN–Transformer hybrid model with attention mechanisms, achieving 91% accuracy.

Tools: CNN-Transformer PCA, KNN, CNN,

Car networking and lane detection of autonomous driving Spring 2023

Investigator (Supervisor: Associate Professor Ronghui Zhang)

- Conducted segmentation based on the network Segnet and lane detection on LaneNet and H-Net
- Compared traditional and novel methods for lane detection, analyzing their advantages and shortcomings respectively

Tools: Deep learning & traditional method on lane detection

Advanced Computer Vision Techniques (Course) Spring 2024

- Stereo vision, SIFT/SURF/ORB feature matching, panorama stitching
- Depth estimation: StereoBM/SGBM and deep stereo (PSMNet)
- 3D reconstruction with NeRF and volumetric rendering

Tools: SIFT/SURF/ORB ,NeRF, StereoBM, PSMNet

C. Robotics Systems, Perception & Control

Novel Embedded Magnetic Field Sensor Arrays for Real-time Shape Estimation of Tendon-driven Continuum Robots 10/2024 - 04/2025

Undergraduate Thesis | Lead Researcher (Supervisor: Professor Mengtang Li)

- Proposed a magnetic sensor array–based shape-sensing method as a low-cost alternative to FBG solutions.
- Designed the robot hardware, sensing module, and calibration procedure.
- Developed kinematic modeling and machine-learning–based angle/shape estimation algorithms.
- Built the experimental platform and validated real-time 3D shape reconstruction.

Gaze-attention based multiple-surgical-tool field of view adjustment for a tendon-driven continuum laparoscopic robot 04/2024 - 08/2024

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Participatory Investigator (Supervisor: Professor Mengtang Li)

- Adjusted the hand-eye consistency errors
- Conducted the simulation experiment for verification

Intelligent cooperative control of mobile robot for industrial warehousing[4] 12/2022 - 12/2023

Project Leader and Principal Investigator (Supervisor: Professor Mengtang Li)

- Established with SolidWorks the model of a smart vehicle with a robotic arm on a Mecanum wheel and completed its simulation and adjustment with Simulink
- Participated in the development and implementation of the path planning algorithm of the vehicle
- Managed the team and made work distribution

D. Advanced Vision Models, Depth Estimation & Diffusion Learning Fall 2025

Depth Estimation & Diffusion Models (Advanced Training, Fall 2025)

- Reproduced DepthPro, EndoDAC, Harvard Rabbit, and DepthCrafter on SCARED; implemented full evaluation pipeline including metric computation and cross-frame point-cloud fusion.
- Built an RGB→Depth diffusion model from scratch.
- Reconstructed DepthCrafter's non-released training pipeline, fine-tuned it on SCARED, and generated stable non-flickering depth videos.
- Investigated model improvements (DINOv2 encoder for stronger contour representation; Flow Matching as a fast alternative to DDIM sampling).

TECHNICAL SKILLS

Programming: Python (PyTorch, NumPy, Pandas, OpenCV), C++
AI / CV: Diffusion Models (DDPM/DDIM/SDE), UNet, Transformers, YOLOv5, DeepSort, DINOv2
Transportation Simulation: SUMO, VISSIM, TransCAD
Engineering Tools: MATLAB/Simulink, SolidWorks, ArcGIS

[4] Established and funded by the College Student Innovation/Entrepreneurship Project